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United States Patent	12403726
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Oizumi; Naoya

Tire

Abstract

In a tire, a main groove is formed in a step shape including circumferential extending portions extending in a circumferential direction at positions where positions in a width direction differ. Inclination directions of a second lug groove and a shoulder lug groove in the circumferential direction with respect to the width direction are opposite to one another. In the second lug grooves, two types of the second lug grooves having different inclination angles with respect to the width direction are alternately disposed. The second lug groove opens to the circumferential extending portion located close to the second lug groove among the circumferential extending portions included in respective center and shoulder main grooves formed in step shapes. A shoulder lug groove opens to the circumferential extending portion located close to the shoulder lug groove among the circumferential extending portions included in the shoulder main groove formed in the step shape.

Inventors:	**Oizumi; Naoya (Kanagawa, JP)**
Applicant:	**The Yokohama Rubber Co., LTD.** (Tokyo, JP)
Family ID:	**78510313**
Assignee:	**The Yokohama Rubber Co., LTD. (Tokyo, JP)**
Appl. No.:	**17/998115**
Filed (or PCT Filed):	**February 25, 2021**
PCT No.:	**PCT/JP2021/007202**
PCT Pub. No.:	**WO2021/229893**
PCT Pub. Date:	November 18, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230173849 A1	Jun. 08, 2023

Foreign Application Priority Data

JP	2020-083900	May. 12, 2020

Publication Classification

Int. Cl.: **B60C11/03** (20060101)

U.S. Cl.:

CPC **B60C11/0306** (20130101); B60C2011/0346 (20130101); B60C2011/0353 (20130101); B60C2011/0365 (20130101); B60C2011/0372 (20130101); B60C2200/14 (20130101)

Field of Classification Search

CPC: B60C (2011/1254); B60C (2011/0374); B60C (2011/0379); B60C (2011/0346); B60C (11/0318)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5269357	12/1992	Killian	152/903	B60C 11/1369
2014/0238563	12/2013	Tanaka	N/A	N/A
2014/0283967	12/2013	Inoue	152/209.18	B60C 11/1236
2016/0089938	12/2015	Iwasaki	152/209.18	B60C 11/1236
2017/0239997	12/2016	Nishino	N/A	N/A
2018/0162172	12/2017	Oji	N/A	B60C 11/03
2018/0272809	12/2017	Kimura	N/A	N/A
2019/0359003	12/2018	Takano	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104002625	12/2013	CN	N/A
110509724	12/2018	CN	N/A
11 2015 003720	12/2016	DE	N/A
2 620 299	12/2014	EP	N/A
3 539 798	12/2018	EP	N/A
64-22601	12/1988	JP	N/A
02179508	12/1989	JP	B60C 11/0309
06183214	12/1993	JP	N/A
2013049325	12/2012	JP	N/A
2013-173521	12/2012	JP	N/A
2014-162439	12/2013	JP	N/A
2017-121919	12/2016	JP	N/A
2017-136998	12/2016	JP	N/A
2017-140927	12/2016	JP	N/A
2017-222280	12/2016	JP	N/A
2018-75880	12/2017	JP	N/A

2018-154280	12/2017	JP	N/A
2019-202608	12/2018	JP	N/A

Primary Examiner: Williams; Cedrick S

Attorney, Agent or Firm: Thorpe North & Western

Background/Summary

TECHNICAL FIELD
(1) The present technology relates to a tire.
BACKGROUND ART
(2) A tire mounted on a vehicle has grooves in a tread surface, for example, to ensure various performances according to a use aspect of the tire, and the performances are improved by devising the shapes of the grooves. For example, in tires described in Japan Unexamined Patent Publication Nos. 2017-121919 A and 2017-136998 A, main grooves extending in a tire circumferential direction are formed in a zigzag shape, and lateral grooves extending in a tire axial direction communicate with the main grooves, thus ensuring on-snow performance, noise performance, and uneven wear resistance performance.
(3) Here, some tires are mounted on a vehicle used for bad roads, and off-road performance as running performance in bad roads is required for the tires. In order to increase off-road performance, methods of increasing a groove area ratio and increasing edge components of grooves are often used. However, increasing the groove area ratio and increasing the edge components of the grooves possibly tend to increase noise generated while a tread contact surface is in contact with a ground. Thus, it has been very difficult to suppress noise while ensuring off-road performance.
SUMMARY
(4) The present technology provides a tire that can provide off-road performance and noise performance in a compatible manner.
(5) A tire according to an embodiment of the present technology includes four main grooves and a plurality of land portions. The four main grooves extend in a tire circumferential direction. The plurality of land portions are defined by the main grooves. The four main grooves include two center main grooves and two shoulder main grooves. The two center main grooves are disposed on both sides of a tire equatorial plane. The two shoulder main grooves are disposed on outer sides in a tire width direction of the two respective center main grooves. Each of the main grooves is formed in a step shape having circumferential extending portions extending in the tire circumferential direction at positions differing in the tire width direction. A second lug groove is disposed between the adjacent center main groove and shoulder main groove. The second lug groove extends in the tire width direction and has both ends opening to the center main groove and the shoulder main groove. A shoulder lug groove is disposed on the outer side in the tire width direction of the shoulder main groove. The shoulder lug groove extends in the tire width direction across a ground contact edge and has one end opening to the shoulder main groove. Inclination directions of the second lug groove and the shoulder lug groove in the tire circumferential direction with respect to the tire width direction are mutually opposite directions. In a plurality of the second lug grooves disposed side by side in the tire circumferential direction, two types of the second lug grooves having different inclination angles with respect to the tire width direction are alternately disposed. The second lug groove opening to the center main groove and the shoulder main groove opens to the circumferential extending portions located close to the second lug groove among the

circumferential extending portions comprised in the respective center main groove and shoulder main groove formed in the step shapes. The shoulder lug groove opening to the shoulder main groove opens to the circumferential extending portion located close to the shoulder lug groove among the circumferential extending portions comprised in the shoulder main groove formed in the step shape.

(6) Additionally, in the tire described above, the main groove preferably has an offset amount Woff in the tire width direction between the circumferential extending portions having the different positions in the tire width direction within a range $0.3 \le (Woff/Wgr) \le 0.8$ with respect to a groove width Wgr of the main groove.

(7) Additionally, in the tire described above, the main groove preferably has a width Ws in the tire width direction of a see-through portion as a portion through which it is seen within a range $1\ \text{mm} < Ws \le 4\ \text{mm}$ in an extension direction view of the main groove.

(8) Additionally, in the tire described above, the second lug groove preferably includes a first second lug groove and a second second lug groove having mutually different inclination angles with respect to the tire width direction and disposed alternately in the tire circumferential direction. The first second lug groove preferably has an angle $\theta\mathbf{1}$ with respect to the tire width direction within a range $15° < \theta 1 \le 35°$. The second second lug groove preferably has an angle $\theta\mathbf{2}$ with respect to the tire width direction within a range $25° \le \theta 2 \le 55°$. The angle $\theta\mathbf{1}$ of the first second lug groove and the angle $\theta\mathbf{2}$ of the second second lug groove with respect to the tire width direction preferably satisfy $\theta\mathbf{1} < \theta\mathbf{2}$.

(9) Additionally, in the tire described above, the first second lug groove preferably has a groove width that widens from an inner side in the tire width direction toward the outer side in the tire width direction. The second second lug groove preferably has a groove width that widens from the outer side in the tire width direction toward the inner side in the tire width direction.

(10) Additionally, in the tire described above, the first second lug groove preferably has a groove depth that deepens from an inner side in the tire width direction toward the outer side in the tire width direction. The second second lug groove preferably has a groove depth that deepens from the outer side in the tire width direction toward the inner side in the tire width direction.

(11) Additionally, in the tire described above, in the center main groove and the shoulder main groove, a relationship between a groove width Wce of the center main groove and a groove width Wsh of the shoulder main groove preferably satisfies $Wce \ge Wsh$.

(12) Additionally, in the tire described above, chamfered portions are preferably formed at edges of the second lug groove and the shoulder lug groove. The chamfered portion preferably has a width We within a range $0.3\ \text{mm} \le We \le 0.8\ \text{mm}$ and a depth De within a range $0.3\ \text{mm} \le De \le 0.8\ \text{mm}$.

(13) Additionally, in the tire described above, a plurality of the land portions preferably include a center land portion disposed between the two center main grooves, a second land portion disposed between the center main groove and the shoulder main groove adjacent in the tire width direction, and a shoulder land portion disposed on the outer side in the tire width direction of the shoulder main groove. In the center land portion, an open sipe having both ends opening to the center main grooves is preferably formed.

(14) Additionally, in the tire described above, semi-closed sipes are preferably formed in the center land portion, the second land portion, and the shoulder land portion. The semi-closed sipes preferably have one end opening to the main grooves and an other end terminating in the land portions.

(15) Additionally, in the tire described above, the semi-closed sipes preferably open to the main grooves with an inclination angle in the tire circumferential direction with respect to the tire width direction at an angle within a range of from 0° or more to 30° or less.

(16) The tire according to the embodiment of the present technology has effects of allowing providing off-road performance and noise performance in a compatible manner.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a tire meridian cross-sectional view illustrating a main portion of a pneumatic tire according to an embodiment.
(2) FIG. **2** is a plan view of a tread portion of the pneumatic tire illustrated in FIG. **1**.
(3) FIG. **3** is a detailed view of a main groove illustrated in FIG. **2**.
(4) FIG. **4** is a detailed view of a portion A of FIG. **2**.
(5) FIG. **5** is a detailed view of the portion A of FIG. **2**, and is an explanatory diagram regarding value specification of second lug grooves.
(6) FIG. **6** is a cross-sectional view along line C-C in FIG. **4**.
(7) FIG. **7** is a detailed view of a portion B of FIG. **2**.
(8) FIGS. **8**A-**8**D are tables showing results of performance evaluation tests of pneumatic tires.
DETAILED DESCRIPTION
(9) Tires according to embodiments of the present technology will be described in detail below with reference to the drawings. However, the present technology is not limited to the embodiment. Constituents of the following embodiments include elements that can be substituted and easily conceived of by a person skilled in the art or that are essentially identical.
(10) Embodiments
(11) In the following description, a description will be given using a pneumatic tire **1** as an example of the tire according to the embodiments of the present technology. The pneumatic tire **1** as an example of the tire can be inflated with any gas including air and inert gas, such as nitrogen.
(12) Additionally, in the following description, the "tire radial direction" refers to a direction orthogonal to a tire rotation axis (not illustrated) which is a rotation axis of the pneumatic tire **1**, the "inner side in the tire radial direction" refers to a side toward the tire rotation axis in the tire radial direction, and the "outer side in the tire radial direction" refers to a side away from the tire rotation axis in the tire radial direction. The term "tire circumferential direction" refers to a circumferential direction with the tire rotation axis as a center axis. Additionally, the term "tire width direction" refers to a direction parallel with the tire rotation axis, the term "inner side in the tire width direction" refers to a side toward a tire equatorial plane (tire equatorial line) CL in the tire width direction, and the term "outer side in the tire width direction" refers to a side away from the tire equatorial plane CL in the tire width direction. The term "tire equatorial plane CL" refers to a plane that is orthogonal to the tire rotation axis and that runs through the center of the tire width of the pneumatic tire **1**. The tire equatorial plane CL aligns, in a position in the tire width direction, with a center line in the tire width direction corresponding to a center position of the pneumatic tire **1** in the tire width direction. The tire width is the width in the tire width direction between portions located on the outermost sides in the tire width direction, or in other words, the distance between the portions that are the most distant from the tire equatorial plane CL in the tire width direction. "Tire equator line" refers to a line in the tire circumferential direction of the pneumatic tire **1** that lies on the tire equatorial plane CL. In the description below, "tire meridian section" refers to a cross-section of the tire taken along a plane that includes the tire rotation axis.
(13) FIG. **1** is a tire meridian cross-sectional view illustrating a main portion of the pneumatic tire **1** according to an embodiment. In the pneumatic tire **1** according to the present embodiment, a tread portion **2** is disposed on a portion on the outermost side in the tire radial direction when viewed in a tire meridian cross-section, and the tread portion **2** includes a tread rubber **4** made of a rubber composition. A surface of the tread portion **2**, that is, a portion in contact with a road surface during traveling of a vehicle (not illustrated) having the pneumatic tires **1** mounted thereon is formed as a tread ground contact surface **3**, and the tread ground contact surface **3** forms a part of a contour of the pneumatic tire **1**. A plurality of main grooves **30** extending in the tire circumferential direction

are formed in the tread ground contact surface **3** in the tread portion **2**, and a plurality of land portions **20** are defined by the plurality of main grooves **30** on the surface of the tread portion **2**.
(14) Shoulder portions **5** are positioned at both ends on outer sides of the tread portion **2** in the tire width direction, and sidewall portions **8** are disposed on inner sides in the tire radial direction of the shoulder portions **5**. In other words, the sidewall portions **8** are disposed on both sides in the tire width direction of the tread portion **2**. In other words, the sidewall portions **8** are disposed at two sections on both sides in the tire width direction of the pneumatic tire **1** and form portions exposed to the outermost sides in the tire width direction of the pneumatic tire **1**.
(15) A bead portion **10** is located on an inner side in the tire radial direction of each of the sidewall portions **8** located on both sides in the tire width direction. Similarly to the sidewall portions **8**, the bead portions **10** are disposed at two sections on both sides of the tire equatorial plane CL. That is, a pair of the bead portions **10** is disposed on both sides in the tire width direction of the tire equatorial plane CL. The bead portions **10** each include a bead core **11**, and a bead filler **12** is provided in the outer side of the bead core **11** in the tire radial direction. The bead core **11** is an annular member formed by bundling bead wires, which are steel wires, and the bead filler **12** is a rubber member disposed in the outer side of the bead core **11** in the tire radial direction.
(16) A belt layer **14** is disposed in the tread portion **2**. The belt layer **14** is formed by a multilayer structure in which a plurality of belts **141**, **142** and a belt cover **143** are layered, and the two layers of the belts **141**, **142** are layered in the present embodiment. The belts **141**, **142** constituting the belt layer **14** are formed by rolling and covering, with coating rubber, a plurality of belt cords made of steel or an organic fiber material, such as polyester, rayon, or nylon, and a belt angle defined as an inclination angle of the belt cords with respect to the tire circumferential direction is within a predetermined range (for example, of 20° or more and 55° or less). Furthermore, the belt angles of the two layers of the belts **141**, **142** differ from each another. Accordingly, the belt layer **14** is configured as a so-called crossply structure in which the two layers of the belts **141**, **142** are layered with the inclination directions of the belt cords intersecting with each another. In other words, the two layers of the belts **141**, **142** are provided as so-called cross belts in which the belt cords provided with the respective belts **141**, **142** are disposed in mutually intersecting orientations.
(17) The belt cover **143** is formed by rolling and covering, with coating rubber, a plurality of belt cover cords made from steel or an organic fiber material, such as polyester, rayon, or nylon, and a belt angle defined as an inclination angle of the belt cover cords with respect to the tire circumferential direction is within a predetermined range (for example, from 0° or more to 10° or less). Additionally, the belt cover **143** is, for example, a strip material formed by coating one or a plurality of belt cover cords with a coating rubber, where the strip material is formed by winding the strip material spirally around the tire rotation axis from the outer side in the tire radial direction of the two layers of the belts **141**, **142**.
(18) A carcass layer **13** containing the cords of radial plies is continuously provided on an inner side in the tire radial direction of the belt layer **14** and on a side of the sidewall portion **8** close to the tire equatorial plane CL. Accordingly, the pneumatic tire **1** according to the present embodiment is configured as a so-called radial tire. The carcass layer **13** has a single layer structure made of one carcass ply or a multilayer structure made of a plurality of carcass plies, and spans between the pair of bead portions **10** disposed on both sides in the tire width direction in a toroidal shape to form a framework of the tire.
(19) Specifically, the carcass layer **13** is disposed from one to the other of the pair of bead portions **10** located on both sides in the tire width direction and is turned back toward the outer side in the tire width direction along the bead cores **11** at the bead portions **10**, wrapping around the bead cores **11** and the bead fillers **12**. The bead filler **12** is a rubber member disposed in a space in the outer side of the bead core **11** in the tire radial direction, the space being formed by folding the carcass layer **13** back at the bead portion **10**. Moreover, the belt layer **14** is disposed on the outer side in the tire radial direction of a portion, located in the tread portion **2**, of the carcass layer **13** spanning

between the pair of bead portions **10**. The carcass ply of the carcass layer **13** is made by coating, with coating rubber, and rolling a plurality of carcass cords made from steel or an organic fiber material such as aramid, nylon, polyester, or rayon. The plurality of carcass cords forming the carcass ply is disposed in parallel at an angle in the tire circumferential direction, the angle with respect to the tire circumferential direction being along a tire meridian direction.
(20) At the bead portion **10**, a rim cushion rubber **17** is disposed on an inner side in the tire radial direction and an outer side in the tire width direction of the bead core **11** and a turned back portion of the carcass layer **13**, the rim cushion rubber **17** forming a contact surface of the bead portion **10** against the rim flange. Additionally, an innerliner **16** is formed along the carcass layer **13** on the inner side of the carcass layer **13** or on the inner portion side of the carcass layer **13** in the pneumatic tire **1**. The innerliner **16** forms a tire inner surface **18** that is a surface on the inner side of the pneumatic tire **1**.
(21) FIG. **2** is a plan view of the tread portion **2** of the pneumatic tire **1** illustrated in FIG. **1**. The four main grooves **30** are formed in the tread portion **2** in the present embodiment. Among the four main grooves **30**, the two main grooves **30** are disposed on each of both sides in the tire width direction of the tire equatorial plane CL. Thus, the four main grooves **30** are formed in the tire width direction side by side. In other words, the four main grooves **30** in total are formed in the tread portion **2**, including: two center main grooves **31** disposed on both sides of the tire equatorial plane CL; and two shoulder main grooves **32** disposed on an outer side in the tire width direction of the two respective center main grooves **31**.
(22) Note that “main groove **30**” refers to a vertical groove in which at least a part is extending in the tire circumferential direction. In general, the main groove **30** has a groove width 3 mm or more and a groove depth of 6 mm or more and has a tread wear indicator (slip sign) therein, indicating terminal stages of wear. In the present embodiment, the main groove **30** has a groove width of 5 mm or more and 8 mm or less and a groove depth of 8 mm or more and 9 mm or less, and an extension direction is substantially parallel to a tire equator line (centerline) where the tire equatorial plane CL and the tread ground contact surface **3** intersect.
(23) The groove width is measured as the maximum value of a distance between groove walls opposed to each other in a groove opening portion when the tire is mounted on a specified rim, inflated to a specified internal pressure, and in an unloaded state. In a configuration in which the land portion includes a notch portion or a chamfered portion on an edge portion thereof, the groove width is measured with intersection points between the tread ground contact surface **3** and extension lines of the groove walls as measurement points, in a cross-sectional view with the groove length direction as a normal line direction. In a configuration in which the grooves extend in a zigzag shape or a wave shape in the tire circumferential direction, the groove width is measured with reference to the center line of the oscillation of the groove walls as measurement points.
(24) The groove depth is measured as the maximum value of a distance from the tread ground contact surface **3** to the groove bottom when the tire is mounted on a specified rim, inflated to the specified internal pressure, and in an unloaded state. Additionally, in a configuration in which the grooves include a partially uneven portion or sipe on the groove bottom, the groove depth is measured excluding these portions.
(25) The plurality of land portions **20** defined by the main grooves **30** include a center land portion **21**, second land portions **22**, and shoulder land portions **23**. Among them, the center land portion **21** is disposed between the two center main grooves **31**, is located on the tire equatorial plane CL, and has both sides in the tire width direction defined by the center main grooves **31**. Additionally, the second land portion **22** is located between the center main groove **31** and the shoulder main groove **32** adjacent in the tire width direction, is disposed on the outer side in the tire width direction of the center land portion **21**, and has both sides in the tire width direction defined by the center main groove **31** and the shoulder main groove **32**. Further, the shoulder land portion **23** is disposed on

the outer side in the tire width direction of the shoulder main groove **32** and is disposed adjacent to the second land portion **22** via the shoulder main groove **32**, and the inner side in the tire width direction of the shoulder land portion **23** is defined by the shoulder main groove **32**.

(26) Additionally, a plurality of lug grooves **40** extending in the tire width direction are formed in the tread ground contact surface **3** of the tread portion **2**, and the lug grooves **40** include second lug grooves **41** and shoulder lug grooves **45**. In the present embodiment, the lug groove **40** has a groove width within the range of from 1.6 mm or more to 6 mm or less, and a groove depth within the range of from 4 mm or more to 7 mm or less. Among the second lug groove **41** and the shoulder lug groove **45**, the second lug groove **41** is disposed between the adjacent center main groove **31** and shoulder main groove **32**, extends in the tire width direction, and has both ends opening to the center main groove **31** and the shoulder main groove **32**. Additionally, the shoulder lug groove **45** is disposed on the outer side in the tire width direction of the shoulder main groove **32**, extends in the tire width direction across a ground contact edge T, and has one end opening to the shoulder main groove **32**.

(27) The ground contact edge T here refers to both outermost edges of a region contacting a flat plate on the tread ground contact surface **3** in the tire width direction when the pneumatic tire **1** is mounted on a specified rim, inflated to a specified internal pressure, placed perpendicular to the flat plate in a stationary state, and loaded with a load corresponding to a specified load and continues in the tire circumferential direction. Here, the specified rim refers to a “standard rim” defined by the Japan Automobile Tyre Manufacturers Association Inc. (JATMA), a “Design Rim” defined by the Tire and Rim Association, Inc. (TRA), or a “Measuring Rim” defined by the European Tyre and Rim Technical Organisation (ETRTO). Moreover, the specified internal pressure refers to a “maximum air pressure” defined by JATMA, a maximum value in “TIRE LOAD LIMITS AT VARIOUS COLD INFLATION PRESSURES” defined by TRA, or “INFLATION PRESSURES” defined by ETRTO. The specified load refers to a “maximum load capacity” defined by JATMA, a maximum value in “TIRE LOAD LIMITS AT VARIOUS COLD INFLATION PRESSURES” defined by TRA, or a “LOAD CAPACITY” defined by ETRTO.

(28) Among the land portions **20** defined by the main grooves **30**, the second land portion **22** having both sides in the tire width direction defined by the center main groove **31** and the shoulder main groove **32** has both sides in the tire circumferential direction defined by the second lug grooves **41** adjacent in the tire circumferential direction. Since both ends of the second lug groove **41** open to the center main groove **31** and the shoulder main groove **32**, the second land portion **22** having both sides in the tire circumferential direction defined by the second lug grooves **41** is formed as the block-shaped land portion **20**.

(29) Additionally, the shoulder land portion **23** defined by the shoulder main groove **32** at the inner side in the tire width direction has both sides in the tire circumferential direction defined by the shoulder lug grooves **45** adjacent in the tire circumferential direction. Since the shoulder lug groove **45** extends in the tire width direction across the ground contact edge T, the shoulder land portion **23** having both sides in the tire circumferential direction defined by the shoulder lug grooves **45** is formed in a block shape at least the portion on the inner side in the tire width direction of the ground contact edge T.

(30) On the other hand, the center land portion **21** having both sides in the tire width direction defined by the center main grooves **31** is formed as the rib-like land portion **20** in which the land portion **20** is formed continuously in the tire circumferential direction without being divided by the lug grooves **40**.

(31) FIG. **3** is a detailed view of the main groove **30** illustrated in FIG. **2**. The four main grooves **30**, the two center main grooves **31** and the two shoulder main grooves **32**, are disposed in the tread portion **2**. The respective main grooves **30** are formed in step shapes including circumferential extending portions **34** extending in the tire circumferential direction at positions where positions in the tire width direction differ. The step shape here refers to a shape in which a center line indicating

a center in a groove width direction forms a rectangular wave or a trapezoidal wave to oscillate. In other words, the main groove **30** includes the two types of the circumferential extending portions **34** having mutually different positions in the tire width direction. The two types of the circumferential extending portions **34** are disposed alternately in the tire circumferential direction, and end portions of the different circumferential extending portions **34** adjacent in the tire circumferential direction are connected by a connection portion **35**. The connection portion **35** also constitutes the main groove **30**.
(32) The two types of the circumferential extending portions **34** include, for example, a plurality of inner circumferential extending portions **34***i* and a plurality of outer circumferential extending portions **34***o* located on the outer side in the tire width direction of the inner circumferential extending portion **34***i*, and the inner circumferential extending portion **34***i* and the outer circumferential extending portion **34***o* are disposed alternately in the tire circumferential direction. In the inner circumferential extending portion **34***i* and the outer circumferential extending portion **34***o*, the inner circumferential extending portions **34***i* have the same position at positions in the tire width direction, and the outer circumferential extending portions **34***o* have the same position at positions in the tire width direction.
(33) Additionally, the connection portion **35** connects the approaching end portions of the inner circumferential extending portion **34***i* and the outer circumferential extending portion **34***o* adjacent in the tire circumferential direction. In the present embodiment, the connection portion **35** is formed so as to extend and be inclined in the tire circumferential direction with respect to the tire width direction. Thus, in the present embodiment, the main groove **30** formed in the step shape is formed in the so-called trapezoidal wave shape.
(34) Thus, in the main groove **30** including the inner circumferential extending portions **34***i*, the outer circumferential extending portions **34***o*, and the connection portions **35**, in one main groove **30**, groove widths Wgr of the inner circumferential extending portion **34***i*, the outer circumferential extending portion **34***o*, and the connection portion **35** are substantially constant.
(35) Thus, the main groove **30** formed in the step shape by the circumferential extending portions **34** and the connection portions **35** has an offset amount Woff in the tire width direction between the circumferential extending portions **34** having the different positions in the tire width direction within the range $0.3 \leq (Woff/Wgr) \leq 0.8$ with respect to the groove width Wgr of the main groove **30**. The offset amount Woff in this case is, for example, a distance in the tire width direction between edges on the same side in the tire width direction of the inner circumferential extending portion **34***i* and the outer circumferential extending portion **34***o*.
(36) Additionally, the main groove **30** has a width Ws in the tire width direction of a see-through portion **37** as a portion through which it is seen within the range 1 mm$<$Ws$\leq$4 mm in the extension direction view of the main groove **30**. In this case, when viewing the inside of the main groove **30** in the extension direction of the main groove **30**, the see-through portion **37** is a portion that can be seen through without being blocked by a groove wall forming the main groove **30**, and the width Ws of the see-through portion **37** is the width in the tire width direction of the portion that can be seen through in this manner.
(37) FIG. **4** is a detailed view of a portion A of FIG. **2**. The four main grooves **30**, the two center main grooves **31** and the two shoulder main grooves **32**, are formed in the tread portion **2**. The center main groove **31** and the shoulder main groove **32** have different groove widths, and the center main groove **31** has the groove width equal to or more than the groove width of the shoulder main groove **32**. That is, in the center main groove **31** and the shoulder main groove **32**, the relationship between a groove width Wce of the center main groove **31** and a groove width Wsh of the shoulder main groove **32** satisfies $Wce \geq Wsh$. Note that the relationship between the groove width Wce of the center main groove **31** and the groove width Wsh of the shoulder main groove **32** is preferably within the range $0.6 \leq (Wsh/Wce) \leq 0.9$.
(38) The lug groove **40** opens to the main groove **30** thus formed. The lug groove **40** opens to the

circumferential extending portion **34** on the side where the lug groove **40** opening to the main groove **30** is located in the tire width direction among the two types of the circumferential extending portions **34** included in the main groove **30**. For example, the second lug groove **41** opening to the center main groove **31** and the shoulder main groove **32** opens to the circumferential extending portion **34** located close to the second lug groove **41** among the circumferential extending portions **34** included in the respective center main groove **31** and shoulder main groove **32** formed in the step shapes. In other words, with respect to the center main groove **31**, since the second lug groove **41** opens to the center main groove **31** from the outer side in the tire width direction, the second lug groove **41** opens to the outer circumferential extending portion **34***o* included in the center main groove **31**. In addition, with respect to the shoulder main groove **32**, since the second lug groove **41** opens to the shoulder main groove **32** from the inner side in the tire width direction, the second lug groove **41** opens to the inner circumferential extending portion **34***i* included in the shoulder main groove **32**.
(39) Also, the shoulder lug groove **45** opening to the shoulder main groove **32** opens to the circumferential extending portion **34** located close to the shoulder lug groove **45** among the circumferential extending portions **34** included in the shoulder main groove **32** formed in the step shape. That is, with respect to the shoulder main groove **32**, since the shoulder lug groove **45** opens to the shoulder main groove **32** from the outer side in the tire width direction, the shoulder lug groove **45** opens to the outer circumferential extending portion **34***o* included in the shoulder main groove **32**. The second lug groove **41** and the shoulder lug groove **45** opening from the inner side and the outer side in the tire width direction of the shoulder main groove **32** with respect to the shoulder main groove **32** open to the different circumferential extending portions **34**, and thus open at different positions in the tire circumferential direction with respect to the shoulder main groove **32**.
(40) Thus, each of the second lug groove **41** and the shoulder lug groove **45** opening to the main groove **30** is formed to be inclined in the tire circumferential direction with respect to the tire width direction while extending in the tire width direction. Then, the inclination directions of the second lug groove **41** and the shoulder lug groove **45** in the tire circumferential direction with respect to the tire width direction are directions opposite to one another. For example, when the second lug groove **41** and the shoulder lug groove **45** extend from the inner side in the tire width direction toward the outer side in the tire width direction, both of them are formed to be inclined in the tire circumferential direction while extending in the tire width direction, but the directions of the second lug groove **41** and the shoulder lug groove **45** toward the tire circumferential direction are directions opposite to one another.
(41) In addition, in the plurality of second lug grooves **41** disposed side by side in the tire circumferential direction, the two types of the second lug grooves **41** having different inclination angles with respect to the tire width direction are alternately disposed. That is, the second lug grooves **41** include a first second lug groove **42** and a second second lug groove **43** having mutually different inclination angles with respect to the tire width direction and disposed alternately in the tire circumferential direction. Thus, in the second land portion **22** having both sides in the tire circumferential direction defined by the second lug grooves **41** adjacent in the tire circumferential direction has both sides in the tire circumferential direction defined by the first second lug groove **42** and the second second lug groove **43** adjacent in the tire circumferential direction.
(42) FIG. **5** is a detailed view of the portion A of FIG. **2**, and is an explanatory diagram regarding value specification of the second lug grooves **41**. Among the first second lug groove **42** and the second second lug groove **43**, the first second lug groove **42** has an angle θ**1** of an inclination in the tire circumferential direction with respect to the tire width direction within the range $15°<θ1≤35°$. The second second lug groove **43** has an angle θ**2** of an inclination in the tire circumferential direction with respect to the tire width direction within the range 25°≤θ**2**≤55. Furthermore, the

angle θ**1** of the first second lug groove **42** and the angle θ**2** of the second second lug groove **43** with respect to the tire width direction satisfy θ**1**<θ**2**. That is, the second second lug groove **43** has the inclination angle in the tire circumferential direction with respect to the tire width direction larger than that of the first second lug groove **42**.

(43) Note that in this case, the inclination angles θ**1**, θ**2** of the first second lug groove **42** and the second second lug groove **43** are angles of straight lines Cg connecting centers of groove widths at end portions on both sides in the tire width direction of the first second lug groove **42** and the second second lug groove **43**.

(44) The first second lug groove **42** and the second second lug groove **43** are formed so as to change the groove widths depending on the positions in the tire width direction. Specifically, the first second lug groove **42** has the monotonically increasing groove width from the inner side in the tire width direction toward the outer side in the tire width direction. That is, as illustrated in FIG. **4**, the first second lug groove **42** has a width narrowed portion **42***n* located close to the center main groove **31** and a width widened portion **42***w* located close to the shoulder main groove **32**, and the width widened portion **42***w* has a groove width wider than that of the width narrowed portion **42***n*. In this way, the first second lug groove **42** has the width widened portion **42***w* having the groove width wider than that of the width narrowed portion **42***n* on the outer side in the tire width direction of the width narrowed portion **42***n*, and thus the first second lug groove **42** has the groove width that widens from the inner side in the tire width direction toward the outer side in the tire width direction.

(45) Conversely, the second second lug groove **43** has the monotonically increasing groove width from the outer side in the tire width direction toward the inner side in the tire width direction. That is, the second second lug groove **43** has a width narrowed portion **43***n* located close to the shoulder main groove **32** and a width widened portion **43***w* located close to the center main groove **31**, and the width widened portion **43***w* has a groove width wider than that of the width narrowed portion **43***n*. In this way, the second second lug groove **43** has the width widened portion **43***w* having the groove width wider than that of the width narrowed portion **43***n* on the inner side in the tire width direction of the width narrowed portion **43***n*, and thus the second second lug groove **43** has the groove width that widens from the outer side in the tire width direction toward the inner side in the tire width direction.

(46) Since the first second lug groove **42** and the second second lug groove **43** disposed alternately in the tire circumferential direction are thus formed, the ways of change in groove widths with respect to the tire width direction are in directions opposite to one another between the first second lug groove **42** and the second second lug groove **43**. In other words, the first second lug groove **42** has the groove width that widens from the tire equatorial plane CL side toward the ground contact edge T side, and the second second lug groove **43** has the groove width that widens from the ground contact edge T side toward the tire equatorial plane CL side, and thus the ways of change in groove widths with respect to the tire width direction are in the directions opposite to one another.

(47) Further, the first second lug groove **42** and the second second lug groove **43** are formed such that the groove depths change depending on the positions in the tire width direction. Specifically, in the first second lug groove **42**, the groove depth at the position of the width widened portion **42***w*, which is disposed on the outer side in the tire width direction of the width narrowed portion **42***n*, is deeper than the groove depth at the position of the width narrowed portion **42***n*. Thus, the first second lug groove **42** has the groove depth that deepens from the inner side in the tire width direction toward the outer side in the tire width direction.

(48) Also, in the second second lug groove **43**, the groove depth at the position of the width widened portion **43***w*, which is disposed on the inner side in the tire width direction of the width narrowed portion **43***n*, is deeper than the groove depth at the position of the width narrowed portion **43***n*. Thus, the second second lug groove **43** has the groove depth that deepens from the outer side in the tire width direction toward the inner side in the tire width direction. Thus, in the first second

lug groove **42** and the second second lug groove **43**, the ways of change in groove depths with respect to the tire width direction are in the directions opposite to one another.

(49) Both of the center main groove **31** and the shoulder main groove **32** to which the second lug groove **41** opens are formed in the step shape due to the inner circumferential extending portion **34***i* and the outer circumferential extending portion **34***o* being alternately disposed in the tire circumferential direction. A relationship between a wavelength λ (FIG. **5**) of the step shapes of the main grooves **30** and a pitch length P (FIG. **5**) of the second lug grooves **41** is within the range $0.4 \leq (\lambda/P) \leq 0.6$.

(50) In this case, the wavelength λ of the step shape of the main groove **30** is the length in the tire circumferential direction of one cycle of the step shape formed by disposing the inner circumferential extending portions **34***i* and the outer circumferential extending portions **34***o* alternately in the tire circumferential direction. For example, the wavelength λ of the step shape of the main groove **30** is a distance in the tire circumferential direction between the end portions on the same side in the tire circumferential direction of the inner circumferential extending portions **34***i* adjacent in the tire circumferential direction, and a distance in the tire circumferential direction between the end portions on the same side in the tire circumferential direction of the outer circumferential extending portions **34***o* adjacent in the tire circumferential direction.

(51) Also, the pitch length P of the second lug grooves **41** is a distance in the tire circumferential direction between the first second lug grooves **42** adjacent via the second second lug groove **43** or between the second second lug grooves **43** adjacent via the first second lug groove **42** in the first second lug grooves **42** and the second second lug grooves **43** disposed alternately in the tire circumferential direction.

(52) Also, sipes **50** are formed in each of the land portions **20** defined by the main grooves **30** and the lug grooves **40** (see FIGS. **4** and **7**). The sipes **50** described herein are formed in a narrow groove shape in the tread ground contact surface **3**. In the sipes **50**, when the pneumatic tire **1** is mounted on a specified rim, inflated in an internal pressure condition at a specified internal pressure, and in an unloaded state, wall surfaces constituting the narrow groove do not contact one another, whereas in a case where the narrow groove is located in a portion of the ground contact surface formed on a flat plate in response to application of a load on the flat plate in the vertical direction or in a case where the land portion **20** including the narrow groove flexes, the wall surfaces constituting the narrow groove or at least parts of portions provided on the wall surface contact one another due to deformation of the land portion **20**. In the present embodiment, the sipe **50** has a groove width of 1.5 mm or less, and a maximum depth from the tread ground contact surface **3** is within the range of from 5 mm or more to 8 mm or less.

(53) The sipes **50** are formed in a configuration of an open sipe **51** (see FIG. **7**) having both ends in a length direction opening to the main groove **30**, a semi-closed sipe **52** (see FIGS. **4** and **7**) having one end opening to the main groove **30** and the other end terminating in the land portion **20**, and a closed sipe **53** (see FIG. **4**) having both ends terminating in the land portion **20**. For example, the open sipes **51** having both ends opening to the center main grooves **31** are formed in the center land portion **21** (see FIG. **7**). Additionally, in the center land portion **21**, the second land portion **22**, and the shoulder land portion **23**, the semi-closed sipes **52** having one end opening to the main grooves **30** and the other end terminating in the land portions **20** are formed (see FIGS. **4** and **7**). Furthermore, the shoulder land portion **23** includes the closed sipes **53** having both ends terminating in the shoulder land portion **23** (see FIG. **4**).

(54) FIG. **6** is a cross-sectional view along line C-C in FIG. **4**. A chamfered portion **48** is formed at an edge of an opening portion to the tread ground contact surface **3** in each of the lug grooves **40**. In other words, in the second lug groove **41** and the shoulder lug groove **45**, the chamfered portions **48** are formed at the edges of the portions defined by the second lug groove **41** and the shoulder lug groove **45** in the land portions **20**. The chamfered portions **48** formed at the edges of the opening portions of the second lug groove **41** and the shoulder lug groove **45** have a width We within the

range 0.3 mm≤We≤0.8 mm and a depth De within the range 0.3 mm≤De≤0.8 mm. In this case, the width We of the chamfered portion **48** is the width of the chamfered portion **48** in the groove width direction of the second lug groove **41** or the shoulder lug groove **45**. Also, the depth De of the chamfered portion **48** is the depth of the chamfered portion **48** in the direction of the groove depth of the second lug groove **41** or the shoulder lug groove **45**.
(55) FIG. **7** is a detailed view of a portion B of FIG. **2**. The open sipe **51** formed in the center land portion **21** extends in the tire width direction across both ends in the tire width direction of the center land portion **21** and has both ends opening to the respective inner circumferential extending portions **34***i* of the center main grooves **31**. Note that in the present embodiment, the open sipe **51** formed in the center land portion **21** extends in the tire width direction while bending in the tire circumferential direction at two locations.
(56) The semi-closed sipes **52** formed in the center land portion **21** include the semi-closed sipe **52** opening to one center main groove **31** and the semi-closed sipe **52** opening to the other center main groove **31** among the two center main grooves **31** that define both sides in the tire width direction of the center land portion **21**. Thus, the semi-closed sipe **52** formed in the center land portion **21** is formed so as to have one end opening to the center main groove **31** and the other end terminating in the center land portion **21**. Similar to the open sipe **51**, the end portion on the side opening to the center main groove **31** of the semi-closed sipe **52**, which has one end opening to the center main groove **31**, opens to the inner circumferential extending portion **34***i* of the center main groove **31**.
(57) In addition, in the semi-closed sipes **52** formed in the center land portion **21**, when the two semi-closed sipes **52** located between the open sipes **51** adjacent in the tire circumferential direction and opening to the different center main grooves **31** are defined as one set, the open sipe **51** and one set of the semi-closed sipes **52** are disposed alternately in the tire circumferential direction. Additionally, one set of the semi-closed sipes **52** formed in the center land portion **21**, that is, the two semi-closed sipes **52** adjacent in the tire circumferential direction, are formed so as not to have portions where the positions in the tire width direction are the same positions. That is, the two semi-closed sipes **52** adjacent in the tire circumferential direction are formed without an overlap in the tire circumferential direction.
(58) In addition, in the open sipe **51** and the semi-closed sipe **52** thus formed in the center land portion **21**, a distance Ls in the tire circumferential direction between the open sipe **51** and the semi-closed sipe **52** mutually disposed at the closest positions is within the range 0.5≤(Ls/Wb)≤1.2 with respect to a ground contact width Wb of the center land portion **21**. In this case, the ground contact width Wb of the center land portion **21** is a maximum width in the tire width direction of the center land portion **21**, and specifically, is a distance in the tire width direction between the edges on the sides of defining the center land portion **21** of the outer circumferential extending portions **34***o* of the center main grooves **31** that define both sides in the tire width direction of the center land portion **21**. Also, the distance Ls in the tire circumferential direction between the open sipe **51** and the semi-closed sipe **52** disposed at the positions closest to one another is the distance in the tire circumferential direction between the opening portion to the center main groove **31** of the semi-closed sipe **52** and the opening portion to the same center main groove **31** of the open sipe **51**.
(59) Also, in the semi-closed sipes **52** (FIG. **4**) formed in the second land portion **22**, the semi-closed sipe **52** opening to the center main groove **31** and the semi-closed sipe **52** opening to the shoulder main groove **32** are disposed between the first second lug groove **42** and the second second lug groove **43** that define both sides in the tire circumferential direction of the second land portion **22**. In other words, in the semi-closed sipes **52** formed in the second land portion **22**, one semi-closed sipe **52** opening to the center main groove **31** and one semi-closed sipe **52** opening to the shoulder main groove **32** are disposed in each of the second land portions **22** formed in the block shape.
(60) Among them, the semi-closed sipe **52** opening to the center main groove **31** and formed in the second land portion **22** has one end opening to the outer circumferential extending portion **34***o* in

the center main groove **31**, and the other end terminating in the second land portion **22**. Additionally, the semi-closed sipe **52** opening to the shoulder main groove **32** and formed in the second land portion **22** has one end opening to the inner circumferential extending portion **34***i* in the shoulder main groove **32**, and the other end terminating in the second land portion **22**.

(61) Additionally, the respective semi-closed sipe **52** opening to the center main groove **31** and formed in the second land portion **22** and semi-closed sipe **52** opening to the shoulder main groove **32** and formed in the second land portion **22** are formed so as not to have portions where the positions in the tire width direction are the same positions. That is, the two semi-closed sipes **52** formed in one second land portion **22** and opening to the different main grooves **30** are formed without an overlap in the tire circumferential direction.

(62) Additionally, the two semi-closed sipes **52** (FIG. **4**) formed in the shoulder land portion **23** are disposed between the shoulder lug grooves **45** adjacent in the tire circumferential direction, and the two semi-closed sipes **52** have one end opening to the shoulder main groove **32** and the other end terminating in the shoulder land portion **23**. Among the two semi-closed sipes **52** disposed between the shoulder lug grooves **45**, one semi-closed sipe **52** opens to the outer circumferential extending portion **34***o* in the shoulder main groove **32**, and the other semi-closed sipe **52** opens to the inner circumferential extending portion **34***i* in the shoulder main groove **32**.

(63) Thus, the semi-closed sipes **52** formed in the respective land portions **20** of the center land portion **21**, the second land portion **22**, and the shoulder land portions **23** open to the main grooves **30** with the inclination angle in the tire circumferential direction with respect to the tire width direction at an angle within the range of from 0° or more to 30° or less. In the present embodiment, among the semi-closed sipes **52**, the semi-closed sipes **52** (FIG. **7**) opening to the center main grooves **31** different from one another and formed in the center land portion **21** have the same inclination direction in the tire circumferential direction with respect to the tire width direction and the inclination angles of substantially the same size.

(64) Additionally, in the semi-closed sipes **52** (FIG. **4**) formed in the second land portion **22**, both of the semi-closed sipe **52** opening to the center main groove **31** and the semi-closed sipe **52** opening to the shoulder main groove **32** have the inclination angles in the tire circumferential direction with respect to the tire width direction of approximately 0°.

(65) Also, in the semi-closed sipes **52** (FIG. **4**) formed in the shoulder land portion **23**, both of the two semi-closed sipes **52** disposed between the shoulder lug grooves **45** adjacent in the tire circumferential direction open with respect to the shoulder lug grooves **45** with the inclination angle in the circumferential direction with respect to the tire width direction of approximately 0°. Additionally, among the semi-closed sipes **52** formed in the shoulder land portion **23** and opening to the shoulder lug groove **45**, the semi-closed sipe **52** on the side opening to the inner circumferential extending portion **34***i* of the shoulder lug groove **45** is bent at a position in the middle in the length direction of the semi-closed sipe **52**.

(66) Further, the closed sipe **53** (FIG. **4**) formed in the shoulder land portion **23** is formed so as to extend in the tire width direction across the ground contact end T, and is bent in the middle in the length direction of the closed sipe **53**.

(67) In this manner, the respective semi-closed sipes **52** and closed sipes **53** formed in the shoulder land portion **23** are formed so as not to have portions where the positions in the tire width direction are the same positions. That is, the semi-closed sipes **52** and the closed sipes **53** formed in one shoulder land portion **23** are formed without an overlap in the tire circumferential direction.

(68) The pneumatic tire **1** according to the present embodiment is used, for example, by being mounted on a vehicle that probably performs off-road travel. In the event of mounting the pneumatic tire **1** on a vehicle, the pneumatic tire **1** is mounted on a rim wheel and inflated with air inside to an inflated state, and then mounted to the vehicle. When the vehicle on which the pneumatic tires **1** are mounted travels, the pneumatic tires **1** rotate while, in the tread ground contact surface **3** on the tread portion **2**, the portion of the tread ground contact surface **3** located at

the bottom comes into contact with the road surface. When the vehicle on which the pneumatic tires **1** are mounted travels on a dry road surface, the vehicle travels mainly by transferring a driving force and a braking force to the road surface and generating a turning force by friction forces between the tread ground contact surface **3** and the road surface. When the vehicle travels on a wet road surface, the vehicle travels in such a way that water between the tread ground contact surface **3** and the road surface enters grooves, such as the main grooves **30** and the lug grooves **40**, and the sipe **50**, and the water between the tread ground contact surface **3** and the road surface is drained through these grooves. As a result, the tread ground contact surface **3** easily contacts the road surface, and the vehicle can travel by the friction force between the tread ground contact surface **3** and the road surface.

(69) Additionally, although the vehicle possibly travels on a so-called off-road, which is an unpaved road, the second lug grooves **41** and the shoulder lug grooves **45** opening to the main grooves **30** are formed in the tread portion **2**, and further the shoulder lug grooves **45** are formed across the ground contact edge T. Thus, for example, during travel on a road surface having fluid mud on its surface, such as a mud road, the mud entering the second lug groove **41** and the shoulder lug groove **45** can be discharged to the main groove **30**, and the mud entering the shoulder lug groove **45** can be discharged to the outer side in the tire width direction of the ground contact edge T. This makes it possible to increase a contact with the ground of the tread ground contact surface **3**, and traction performance during off-road travel can be ensured.

(70) Additionally, since the main groove **30** is formed in the step shape, a resistance in the tire circumferential direction against the mud that has entered the inside of the main groove **30** during off-road travel, such as a mud road, can be provided. In this way, traction performance during off-road travel can be more reliably ensured.

(71) Also, the second lug grooves **41** and the shoulder lug grooves **45** open to the circumferential extending portions **34** located on the sides where the respective lug grooves **40** open in the tire width direction with respect to the main grooves **30** formed in the step shapes in this manner. Thus, the rigidity of the land portions **20** defined by the main grooves **30** and the lug grooves **40** can be ensured, and steering stability can be ensured regardless of the traveling road surface.

(72) In addition, although there is a road surface with many recesses/protrusions, such as a rocky area, in off-road, a posture of the vehicle becomes unstable on the road surface with many recesses/protrusions, and a ground contact area between the tread ground contact surface **3** and the road surface decreases, and thus, a drive wheel idles when a driving force is transferred in some cases. In this way, in the case where the drive wheel idles during traveling in a rocky area, when a pointed portion of a rock enters the inside of the main groove **30**, the pneumatic tire **1** rotates while the rock keeps contacting the groove bottom of the main groove **30**. In this case, when the main groove **30** extends straight in the tire circumferential direction, the pneumatic tire **1** rotates while keeping the state in which the rock contacts the groove bottom of the main groove **30**, and thus a rubber of the groove bottom of the main groove **30** is damaged by the rock, and possibly causes cracks in the groove bottom of the main groove **30**.

(73) In contrast, in the present embodiment, the main groove **30** is formed in the step shape including the circumferential extending portions **34** extending in the tire circumferential direction at the positions where the positions in the tire width direction differ. Thus, when a pointed portion of a rock enters the main groove **30**, the pointed portion enters the circumferential extending portion **34** of the main groove **30** in many cases. When the drive wheel idles in the state, the rock abuts on the end portion in the tire circumferential direction of the circumferential extending portion **34**. In other words, the rock abuts on the portion defined by the end portions in the tire circumferential direction of the circumferential extending portions **34** in the land portions **20**. As a result, the rock that has entered the main groove **30** is discharged from the main groove **30**. Thus, entrance of a rock into the main groove **30** and rotation of the pneumatic tire **1** in a state where the rock keeps contacting the groove bottom of the main groove **30** can be suppressed. Accordingly,

damage of the rubber of the groove bottom of the main groove **30** by the rock can be suppressed, and cracks in the groove bottom of the main groove **30** can be suppressed.

(74) Additionally, since the main groove **30** is formed in the step shape, formation of a large number of portions where the rigidity of the land portions **20** decrease can be suppressed, as in the case where, for example, the shape of the main groove **30** is formed in a zigzag shape oscillating in the tire width direction while extending in the tire circumferential direction in order to discharge the pointed portion of the rock from the main groove **30**. Thus, the rigidity of the land portions **20** can be ensured, and steering stability can be more reliably ensured regardless of the traveling road surface.

(75) Additionally, since the inclination directions of the second lug groove **41** and the shoulder lug groove **45** are the mutually opposite directions, when a sound generated while the tread ground contact surface **3** contacts the ground is transmitted to the second lug groove **41** and the shoulder lug groove **45** and comes out of the outer side in the tire width direction of the ground contact edge T, the sound can be less likely to transmit between the second lug groove **41** and the shoulder lug groove **45**. This makes it possible to increase noise performance, which is the performance of making it difficult to generate a sound during rotation of the pneumatic tire **1**. Furthermore, in the plurality of second lug grooves **41** disposed side by side in the tire circumferential direction, the two types of the second lug grooves **41** having the different inclination angles with respect to the tire width direction are disposed alternately, and thus, a frequency of a sound generated while the second land portions **22** defined by the second lug grooves **41** contact the ground can be dispersed. Thus, amplification of the sound at a specific frequency and an increase in sound volume can be suppressed, and noise performance can be more reliably enhanced. Consequently, off-road performance and noise performance can be provided in a compatible manner.

(76) In addition, the main groove **30** has the offset amount Woff in the tire width direction between the circumferential extending portions **34** having the different positions in the tire width direction within the range $0.3 \leq (Woff/Wgr) \leq 0.8$ with respect to the groove width Wgr of the main groove **30**. Thus, while wet performance is ensured, cracks in the groove bottom of the main groove **30** can be more reliably suppressed. In other words, when the offset amount Woff between the circumferential extending portions **34** of the main groove **30** is $(Woff/Wgr) < 0.3$ with respect to the groove width Wgr of the main groove **30**, the offset amount Woff between the circumferential extending portions **34** is too small. Thus, when the drive wheel idles with the pointed portion of the rock entering the inside of the main groove **30**, the rock is possibly difficult to be discharged from the main groove **30**. In this case, the pneumatic tire **1** rotates in a state where the rock keeps contacting the groove bottom of the main groove **30**, and cracks in the groove bottom of the main groove **30** are possibly difficult to be suppressed. In addition, when the offset amount Woff between the circumferential extending portions **34** of the main groove **30** is $(Woff/Wgr) > 0.8$ with respect to the groove width Wgr of the main groove **30**, the offset amount Woff between the circumferential extending portions **34** is too large. Thus, when water flows in the main groove **30** during traveling on wet road surfaces, the water is possibly less likely to flow in the main groove **30**. In this case, the water between the tread ground contact surface **3** and the road surface is difficult to be drained, and thus wet performance, which is the running performance on wet road surfaces, is possibly difficult to be ensured.

(77) In contrast, when the offset amount Woff between the circumferential extending portions **34** of the main groove **30** is within the range $0.3 \leq (Woff/Wgr) \leq 0.8$ with respect to the groove width Wgr of the main groove **30**, while ease of flow of water when the water enters the main groove **30** is ensured, a discharge property of a rock when a pointed portion of the rock enters the inside of the main groove **30** can be ensured. Thus, while wet performance is ensured, cracks in the groove bottom of the main groove **30** can be more reliably suppressed. As a result, while wet performance is ensured, durability during off-road travel can be improved.

(78) Additionally, the width Ws in the tire width direction of the see-through portion **37** of the main

groove **30** is within the range $1\ \text{mm} < Ws \leq 4\ \text{mm}$. Thus, while wet performance is ensured, cracks in the groove bottom of the main groove **30** can be more reliably suppressed. In other words, in a case where the width Ws of the see-through portion **37** of the main groove **30** is $Ws \leq 1\ \text{mm}$, the width Ws of the see-through portion **37** is too small, and the ease of flow of water in the main groove **30** decreases. Thus, drainage properties in the main groove **30** decrease, and wet performance is possibly difficult to be ensured. In addition, in a case where the width Ws of the see-through portion **37** of the main groove **30** is $Ws > 4\ \text{mm}$, the width Ws of the see-through portion **37** is too large. Thus, when the pneumatic tire **1** rotates in a state where a pointed portion of a rock enters the main groove **30**, the rock is possibly difficult to be discharged from the main groove **30**. In this case, since the pneumatic tire **1** continues rotating while the pointed portion of the rock contacts the groove bottom of the main groove **30**, cracks in the groove bottom of the main groove **30** are possibly difficult to be suppressed.

(79) In contrast, when the width Ws of the see-through portion **37** of the main groove **30** is within the range $1\ \text{mm} < Ws \leq 4\ \text{mm}$, while ease of flow of water when the water enters the main groove **30** is ensured, a discharge property of a rock when a pointed portion of the rock enters the inside of the main groove **30** can be ensured. Thus, while wet performance is ensured, cracks in the groove bottom of the main groove **30** can be more reliably suppressed. As a result, while wet performance is ensured, durability during off-road travel can be improved.

(80) Additionally, the angle $\theta 1$ of the first second lug groove **42** is within the range $15^\circ < \theta 1 \leq 35^\circ$ and the angle $\theta 2$ of the second second lug groove **43** is within the range $25^\circ \leq \theta 2 \leq 55^\circ$ with respect to the tire width direction, and $\theta 1 < \theta 2$ is satisfied. Accordingly, off-road performance, wet performance, and noise performance can be more reliably provided in a compatible manner, and the durability during off-road travel can be improved. That is, when the angle $\theta 1$ of the first second lug groove **42** is $\theta 1 \leq 15^\circ$, and the angle $\theta 2$ of the second second lug groove **43** is $\theta 2 < 25^\circ$, the angle $\theta 1$ of the first second lug groove **42** and the angle $\theta 2$ of the second second lug groove **43** are too small, and thus the lengths of the first second lug groove **42** and the second second lug groove **43** are possibly difficult to be ensured. Thus, the effects of wet performance and off-road performance by the first second lug groove **42** and the second second lug groove **43** are possibly difficult to be effectively obtained. In addition, in a case where the angle $\theta 1$ of the first second lug groove **42** is $\theta 1 > 35^\circ$ and the angle $\theta 2$ of the second second lug groove **43** is $\theta 2 > 55^\circ$, the angle $\theta 1$ of the first second lug groove **42** and the angle $\theta 2$ of the second second lug groove **43** are too large. Thus, a portion at which the angle becomes too small possibly occurs at the corner portion where the main groove **30** meets the second lug groove **41** in the second land portion **22**. In this case, the portion at which rigidity becomes too small possibly occurs at the corner portion where the main groove **30** meets the second lug groove **41** in the second land portion **22**. Due to the low rigidity, damage, such as separation, possibly occurs in the second land portion **22** during off-road travel, such as a rocky area.

(81) In contrast, when the angle $\theta 1$ of the first second lug groove **42** is within the range $15^\circ < \theta 1 \leq 35^\circ$ and the angle $\theta 2$ of the second second lug groove **43** is within the range $25^\circ \leq \theta 2 \leq 55^\circ$ with respect to the tire width direction, the lengths of the first second lug groove **42** and the second second lug groove **43** can be ensured while too low rigidity at the corner portion where the main grooves **30** intersects with the second lug groove **41** in the second land portion **22** is suppressed. This makes it possible to more reliably improve wet performance and off-road performance while suppressing damage, such as separation, in the second land portion **22**. Furthermore, in a case where the angle $\theta 1$ of the first second lug groove **42** and the angle $\theta 2$ of the second second lug groove **43** with respect to the tire width direction satisfy $\theta 1 < \theta 2$, the lengths of the first second lug groove **42** and the second second lug groove **43** can be differentiated, and thus a frequency of air column resonance generated while the second lug groove **41** contacts the ground can be dispersed. Consequently, off-road performance, wet performance, and noise performance can be more reliably provided in a compatible manner, and the durability during off-road travel can be improved.

(82) Further, the first second lug groove **42** has the groove width that widens from the inner side in the tire width direction to the outer side in the tire width direction, and the second second lug groove **43** has the groove width that widens from the outer side in the tire width direction to the inner side in the tire width direction. Thus, the ways of change in groove widths with respect to the tire width direction are in the directions opposite to one another. As a result, air column resonance in the first second lug groove **42** and air column resonance in the second second lug groove **43** can be differentiated, so that a frequency of a sound generated while the second land portions **22** defined by the first second lug grooves **42** and the second second lug grooves **43** contact the ground can be more reliably dispersed. Thus, amplification of the sound at a specific frequency and an increase in sound volume can be more reliably suppressed. As a result, noise performance can be more reliably improved.
(83) Additionally, the first second lug groove **42** has the groove depth that deepens from the inner side in the tire width direction toward the outer side in the tire width direction, and the second second lug groove **43** has the groove depth that deepens from the outer side in the tire width direction toward the inner side in the tire width direction, and thus the ways of change in groove depths with respect to the tire width direction are in the directions opposite to one another. As a result, air column resonance in the first second lug groove **42** and air column resonance in the second second lug groove **43** can be differentiated, so that a frequency of a sound generated while the second land portions **22** defined by the first second lug grooves **42** and the second second lug grooves **43** contact the ground can be more reliably dispersed. As a result, noise performance can be more reliably improved.
(84) Additionally, the relationship between the groove width Wce of the center main groove **31** and the groove width Wsh of the shoulder main groove **32** satisfies Wce≥Wsh. Accordingly, ease of transmission of a sound in the shoulder main groove **32** can be limited, and emission of a sound while the tread ground contact surface **3** contacts the ground to the outside of the vehicle from the shoulder main groove **32** through the shoulder lug groove **45** can be suppressed. That is, the shoulder main groove **32** is disposed on the outer side in the tire width direction of the center main groove **31**, and one shoulder main groove **32** among the two shoulder main grooves **32** is disposed on the outermost side in a mounting direction when the pneumatic tire **1** is mounted on a vehicle among the plurality of main grooves **30**. Accordingly, when the sound generated while the tread ground contact surface **3** contacts the ground is transmitted to the shoulder main groove **32** disposed on the outermost side in the mounting direction to the vehicle, the sound flows from the shoulder main groove **32** to the shoulder lug groove **45** and is emitted from the shoulder lug groove **45** to the outside of the vehicle, thus being emitted to the outside of the vehicle as a noise. Since the sound emitted outside the vehicle is thus likely to be transmitted to the shoulder main groove **32**, the groove width Wsh of the shoulder main groove **32** is set to be less than or equal to the groove width Wce of the center main groove **31** to limit the ease of transmission of a sound in the shoulder main groove **32**, thus ensuring suppressing the emission of the sound generated while the tread ground contact surface **3** contacts the ground to the outside of the vehicle. Thus, noise performance can be more reliably improved.
(85) Additionally, since the chamfered portions **48** are formed at the edges of the second lug groove **41** and the shoulder lug groove **45**, the durability during off-road travel can be improved. In other words, during off-road travel, by entrance of, for example, pointed portions of rocks into the lug grooves **40** extending in the tire width direction, a large force in the tire circumferential direction acts on the portions defined by the lug grooves **40** in the land portions **20**, and so-called chunking, which is a failure of partial separation of the tread portion **2**, possibly occurs. In contrast, in the present embodiment, the chamfered portions **48** are formed at the edges of the second lug groove **41** and the shoulder lug groove **45**. Thus, the force in the tire circumferential direction acting on the land portion **20** when, for example, a pointed portion of a rock enters the second lug groove **41** or the shoulder lug groove **45** can be alleviated. This makes it possible to suppress partial separation

of the tread portion **2** due to the force acting on the land portion **20** from the rock even when, for example, a pointed portion of a rock enters the second lug groove **41** or the shoulder lug groove **45** during off-road travel, thus ensuring suppressing chunking. As a result, the durability during off-road travel can be improved.

(86) Furthermore, the chamfered portions **48** of the second lug groove **41** and the shoulder lug groove **45** have the widths We within the range 0.3 mm≤We≤0.8 mm, and the depths De within the range 0.3 mm≤De≤0.8 mm, and thus noise performance and the durability during off-road travel can be improved. In other words, when the width We or the depth De of the chamfered portion **48** is less than 0.3 mm, the size of the chamfered portion **48** is too small, so even when the chamfered portions **48** are formed at the edges of the second lug groove **41** and the shoulder lug groove **45**, the force acting on the land portion **20** when, for example, rocks enter the second lug groove **41** and the shoulder lug groove **45** is possibly difficult to be alleviated. In addition, when the width We or the depth De of the chamfered portion **48** is larger than 0.8 mm, the size of the chamfered portion **48** is too large, and thus the ground contact areas of the land portions **20** defined by the second lug grooves **41** and the shoulder lug grooves **45** are possibly too small. In this case, since a surface pressure while the land portion **20** contacts the ground increases, a hitting sound when the land portion **20** contacts the road surface is likely to increase, and noise performance is possibly likely to degrade.

(87) In contrast, when the widths We or the depths De of the chamfered portions **48** of the second lug groove **41** and the shoulder lug groove **45** are within the range of 0.3 mm or more to 0.8 mm or less, the force acting on the land portions **20** when, for example, a rock enters the second lug groove **41** and the shoulder lug groove **45** can be alleviated while the excessively small ground contact area of the land portions **20** defined by the second lug grooves **41** and the shoulder lug grooves **45** is suppressed. As a result, noise performance and the durability during off-road travel can be more reliably improved.

(88) Additionally, the center land portion **21** is not divided in the tire circumferential direction by the lug grooves **40**, and the open sipes **51** having both ends opening to the center main grooves **31** are formed in the center land portion **21**. Accordingly, the running performance in a sandy ground and running performance in a rocky area and a mud road can be provided in a compatible manner. In other words, even when the lug groove **40** is formed to ensure off-road performance, in a sandy ground, sand entering the lug groove **40** is easily scraped up from the sandy ground accompanied by the rotation of the pneumatic tire **1**. As a result, when the driving force from the drive wheel is transferred to the sandy ground, the pneumatic tire **1** digs the sandy ground while scraping up the sand with the lug grooves **40** and sinks into the sandy ground, possibly making transfer of the driving force difficult.

(89) In contrast, in the present embodiment, since the center land portion **21** is not divided by the lug grooves **40**, the ground contact area of the center land portion **21** can be increased, and the ground contact pressure of the tread ground contact surface **3** can be reduced. As a result, even when the driving force from the drive wheel is transferred to a sandy ground, digging of the sandy ground by the pneumatic tire **1** and sinking of the pneumatic tire **1** into the sandy ground can be suppressed. Accordingly, traction performance in a sandy ground can be ensured.

(90) On the other hand, in a case where the center land portion **21** is not separated in the tire circumferential direction by the lug grooves **40**, the rigidity of the center land portion **21** is too high. Accordingly, the center land portion **21** is possibly difficult to follow recesses/protrusions of a road surface during traveling on, for example, a rock area or a mud road. In contrast, in the present embodiment, since the open sipes **51** are formed in the center land portion **21**, the rigidity of the center land portion **21** can be appropriately reduced, and the center land portion **21** can be deformed in accordance with the recesses/protrusions of the road surface during traveling on, for example, a rock area or a mud road. This allows the center land portion **21** to follow the recesses/protrusions of the road surface and allows ensuring traction performance and steering

stability during traveling on, for example, a rocky area and a mud road. Consequently, the running performance in a sandy ground and the running performance in a rocky area and a mud road can be provided in a compatible manner, thus ensuring more reliably improving off-road performance.

(91) Additionally, since the semi-closed sipes **52** are formed in the center land portion **21**, the second land portion **22**, and the shoulder land portion **23**, the rigidity of the land portions **20** can be appropriately reduced, the ground contact pressure can be appropriately reduced, and thus the running performance and the durability during off-road travel can be more reliably improved. In other words, when the rigidity of the land portion **20** is too low, chunking is possibly likely to occur in the tread portion **2** during traveling on a hard road surface having large recesses/protrusions, such as a rocky area. In addition, when the rigidity of the land portion **20** is too high, the land portion **20** is less likely to follow recesses/protrusions of a road surface during traveling on an offroad road surface. Accordingly, running performance during off-road travel is possibly difficult to be ensured. In contrast, in the present embodiment, formation of the semi-closed sipes **52** in the land portions **20** allows appropriately reducing the rigidity of the land portions **20**. Thus, while chunking in the tread portion **2** is suppressed, the land portion **20** can follow the recesses/protrusions of the road surface, and the running performance during off-road travel can be ensured.

(92) In addition, in the case where the lug grooves **40** having one end opening to the main groove **30** and the other end terminating in the land portion **20** are formed in the land portion **20**, the ground contact area of the land portion **20** becomes small and the ground contact pressure increases.

(93) Accordingly, sinking in the sandy ground is possibly likely to occur during travel on the sandy ground. This possibly makes it difficult to ensure traction performance during traveling on a sandy ground. In contrast, in the present embodiment, in the center land portion **21**, the second land portion **22**, and the shoulder land portion **23**, the semi-closed sipes **52** having one end opening to the main grooves **30** and the other end terminating in the land portions **20** are formed, and thus the excessively small ground contact areas of the land portions **20** can be suppressed. Since this allows suppressing the excessively high ground contact pressure of the land portion **20**, digging of the sandy ground by the pneumatic tire **1** and sinking of the pneumatic tire **1** into the sandy ground can be suppressed, and traction performance in a sandy ground can be ensured. Consequently, the running performance and the durability during off-road travel can be more reliably improved.

(94) Additionally, the semi-closed sipe **52** opens to the main groove **30** with the inclination angle in the tire circumferential direction with respect to the tire width direction at the angle within the range of from 0° or more to 30° or less, and thus the traction performance during off-road travel can be more reliably improved. In other words, when the inclination angle of the semi-closed sipe **52** with respect to the tire width direction is larger than 30°, an edge component of the semi-closed sipe **52** with respect to the tire circumferential direction is small, and thus traction performance by the semi-closed sipe **52** in off-road is possibly difficult to be ensured.

(95) In contrast, when the inclination angle of the semi-closed sipe **52** in the tire circumferential direction with respect to the tire width direction is within the range of from 0° or more to 30° or less, the edge component of the semi-closed sipe **52** with respect to the tire circumferential direction can be ensured, and traction performance can be more reliably ensured. In particular, on a rocky area or a mud road, by setting the inclination angle of the semi-closed sipe **52** with respect to the tire width direction to within the range of from 0° or more to 30° or less, when the semi-closed sipes **52** open at the start of moving of the land portion **20**, the edge component of the semi-closed sipes **52** can be more reliably exerted with respect to the tire circumferential direction. This allows improving traction performance during travel on an off-road, such as a rocky area and a mud road, at low speed more reliably. As a result, off-road performance can be more reliably improved.

(96) Modified Examples

(97) Note that in the embodiments described above, in the main groove **30**, the connection portions

35 connecting the circumferential extending portions **34** adjacent in the tire circumferential direction are inclined in the tire circumferential direction with respect to the tire width direction and extend to form the main grooves **30** in the trapezoidal wave shape. However, the main groove **30** may be formed in a shape other than the trapezoidal wave shape. The main groove **30** may be formed in a rectangular wave shape, for example, by forming the connection portions **35** connecting the circumferential extending portions **34** adjacent in the tire circumferential direction so as to extend substantially parallel to the tire width direction. In addition, the plurality of main grooves **30** may differ in the configuration of the step shape between the main grooves **30**. As long as the main groove **30** is formed in the step shape having the circumferential extending portions **34** extending in the tire circumferential direction at the positions where the positions in the tire width direction differ, the detailed shape of the main groove **30** may be any shape.
(98) Additionally, in the embodiment described above, the first second lug groove **42** has the width widened portion **42***w* on the outer side in the tire width direction of the width narrowed portion **42***n*, and thus the groove width widens from the inner side in the tire width direction toward the outer side in the tire width direction, but the first second lug groove **42** may change the groove width in a configuration other than that. The first second lug groove **42** may have a widened groove width from the inner side in the tire width direction toward the outer side in the tire width direction without having a step in the groove width direction like the boundary portion between the width narrowed portion **42***n* and the width widened portion **42***w*. Similarly, the second second lug groove **43** has the width widened portion **43***w* on the inner side in the tire width direction of the width narrowed portion **43***n*, and thus the groove width widens from the outer side in the tire width direction toward the inner side in the tire width direction, but the second second lug groove **43** may also be formed so as not to have a step in the groove width direction and the groove width widens from the outer side in the tire width direction toward the inner side in the tire width direction. The first second lug groove **42** only needs to monotonically increase the groove width from the inner side in the tire width direction toward the outer side in the tire width direction, and the second second lug groove **43** only needs to monotonically increase the groove width from the outer side in the tire width direction toward the inner side in the tire width direction.
(99) In the embodiment described above, although the pneumatic tire **1** is used for description as an example of the tire according to the embodiment of the present technology, the tire according to the embodiment of the present technology may be a tire other than the pneumatic tire **1**. The tire according to the embodiment of the present technology may be, for example, a so-called airless tire that can be used without filling a gas.
EXAMPLES
(100) FIGS. **8**A-**8**D are tables showing results of performance evaluation tests of pneumatic tires. In relation to the pneumatic tire **1** described above, description will be given of performance evaluation tests conducted on a pneumatic tire according to Conventional Example, the pneumatic tires **1** according to embodiments of the present technology, and pneumatic tires according to Comparative Examples to be compared with pneumatic tires **1** according to the embodiments of the present technology. In the performance evaluation test, for noise performance during vehicle travel, off-road performance as running performance during off-road travel, and chunking in the tread portion were tested.
(101) The performance evaluation tests were conducted by mounting the pneumatic tire **1** with a tire nominal size of 285/60R18 116V specified by JATMA on a JATMA standard rim wheel with a rim size of 18×8 J, adjusting the air pressure to 230 kPa, mounting the test tires on the evaluation vehicle of a four-wheel-drive SUS vehicle, and then driving the evaluation vehicle.
(102) In the evaluation method for each test item, for noise performance, noise levels at speeds of 60 km/h and 100 km/h on a flat road were evaluated by sensory evaluation by test driver with Conventional Example 1 described below being assigned as the reference **100**. Largerindex value of noise performance indicates lower noise during travel on a flat road and superior noise

performance.

(103) Also, for off-road performance, running performance during traveling on the respective evaluation road surfaces of a rocky area, a mud road, and a sandy ground was evaluated by sensory evaluation by a test driver, and the total point of the sensory evaluation in the respective evaluation road surfaces was evaluated with Conventional Example 1 described below being assigned as the reference **100**. Larger index value of off-road performance indicates higher total point of running performance during travel on the respective evaluation road surfaces and superior off-road performance.

(104) In addition, for chunking, the tread portion of the test tire after conducting the evaluation test for off-road performance was examined, and the number of clear chunkings having ranges and depths of 5 mm or more on the tire circumference was counted. The average number of chunkings per pitch of the tread pattern was calculated from the number of chunkings on the tire circumference thus counted. When the average number of chunkings per pitch was one or more, it was determined that the chunkings were large. When the average number of chunkings per pitch was less than one, it was determined that the chunkings were small.

(105) The performance evaluation tests were conducted on 16 types of pneumatic tires including pneumatic tires according to Conventional Examples 1 to 3 corresponding to an example of conventional pneumatic tires, Examples 1 to 11 corresponding to the pneumatic tires **1** according to the embodiments of the present technology, and Comparative Examples 1 and 2 corresponding to pneumatic tires to be compared with the pneumatic tires **1** according to the embodiments of the present technology. Among them, in Conventional Example 1, second lug grooves having two types of different inclination angles are not provided, and lug grooves do not open to circumferential extending portions located close to the lug grooves of a main groove. Additionally, in Conventional Example 2, inclination directions of a second lug groove and a shoulder lug groove with respect to a tire width direction are in the mutually same direction, two types of second lug grooves having different inclination angles are not provided, and further lug grooves do not open to circumferential extending portions located close to the lug grooves of a main groove. Additionally, in Conventional Example 3, a main groove does not have a zigzag shape extending in a tire circumferential direction and oscillating in a tire width direction, two types of second lug grooves having different inclination angles are not provided, and further lug grooves do not open to circumferential extending portions located close to the lug grooves of a main groove. In addition, in Comparative Example 1, lug grooves do not open to circumferential extending portions located close to the lug grooves of a main groove. Additionally, in Comparative Example 2, two types of second lug grooves having different inclination angles are not provided.

(106) On the other hand, in Examples 1 to 11, which are examples of the pneumatic tires **1** according to the embodiment of the present technology, all of the main grooves **30** have the step shapes, the inclination directions of the second lug groove **41** and the shoulder lug groove **45** with respect to the tire width direction are in opposite directions, the second lug grooves **41** having the different inclination angles are provided, and the lug grooves **40** open to the circumferential extending portions **34** of the main groove **30** located close to the lug grooves **40**. Furthermore, each of the ratios of the offset amounts Woff between the circumferential extending portions **34** to the groove widths Wgr of the main grooves **30** (Woff/Wgr), the widths of the see-through portions **37** of the main grooves **30**, the angles θ**1** of the first second lug grooves **42** and the angles θ**2** of the second second lug grooves **43** with respect to the tire width directions, the sides where the groove widths of the first second lug grooves **42** widen, the sides where the groove widths of the second second lug grooves **43** widen, the sides where the groove depths of the first second lug grooves **42** deepen, the sides where the groove depths of the second second lug grooves **43** deepen, the ratios of the groove widths Wsh of the shoulder main grooves **32** to the groove widths Wce of the center main grooves **31** (Wsh/Wce), and presence of the chamfered portions **48** of the lug grooves **40** are differentiated.

(107) As a result of conducting the performance evaluation tests using these pneumatic tires **1**, as illustrated in FIGS. **8**A-**8**D, it is found that the pneumatic tires **1** according to Examples 1 to 11 do not decrease in noise performance or off-road performance compared with Conventional Examples 1 to 3 or Comparative Examples 1 and 2, and at least one of the performances can be improved. In other words, the pneumatic tires **1** according to Examples 1 to 11 can provide off-road performance and noise performance in a compatible manner.

Claims

1. A tire, comprising: four main grooves extending in a tire circumferential direction; a plurality of land portions defined by the four main grooves; and the four main grooves comprising two center main grooves and two shoulder main grooves, the two center main grooves being disposed on both sides of a tire equatorial plane, the two shoulder main grooves being disposed on outer sides in a tire width direction of the two respective center main grooves, each of the four main grooves being formed in a step shape having circumferential extending portions extending in the tire circumferential direction at positions differing in the tire width direction, a second lug groove being disposed between one center main groove of the two center main grooves and one shoulder main groove of the two shoulder main grooves which is adjacent to the one center main groove, the second lug groove extending in the tire width direction and having both ends opening to the one center main groove and the one shoulder main groove, a shoulder lug groove being disposed on the outer side in the tire width direction of the one shoulder main groove, the shoulder lug groove extending in the tire width direction across a ground contact edge and having one end opening to the one shoulder main groove, inclination directions of the second lug groove and the shoulder lug groove in the tire circumferential direction with respect to the tire width direction being mutually opposite directions, in a plurality of the second lug grooves disposed side by side in the tire circumferential direction, two types of the second lug grooves having different inclination angles with respect to the tire width direction being alternately disposed; the second lug groove opening to the one center main groove and the one shoulder main groove opening to the circumferential extending portions located close to the second lug groove among the circumferential extending portions comprised in respective of the one center main groove and the one shoulder main groove formed in the step shapes, the shoulder lug groove opening to the one shoulder main groove opening to the circumferential extending portion located close to the shoulder lug groove among the circumferential extending portions comprised in the one shoulder main groove formed in the step shape, the step shape having two types of the circumferential extending portions having mutually different positions in the tire width direction and a connection portion connecting the two types of the circumferential extending portions adjacent in the tire circumferential direction, a center line of the step shape forming a rectangular wave or a trapezoidal wave to oscillate, the connection portion of the shoulder main groove connecting the second lug groove and the shoulder lug groove, the second lug groove comprising a first second lug groove and a second second lug groove having mutually different inclination angles with respect to the tire width direction and disposed alternately in the tire circumferential direction, and a relationship between a wavelength λ of the step shape of the main grooves and a pitch length P of the second lug groove being within a range $0.4 \leq (\lambda/P) \leq 0.6$, the pitch length P of the second lug groove being a distance in the tire circumferential direction between the first second lug grooves adjacent in the tire circumferential direction.

2. A tire, comprising: four main grooves extending in a tire circumferential direction; a plurality of land portions defined by the four main grooves; and the four main grooves comprising two center main grooves and two shoulder main grooves, the two center main grooves being disposed on both sides of a tire equatorial plane, the two shoulder main grooves being disposed on outer sides in a tire width direction of the two respective center main grooves, each of the four main grooves being

formed in a step shape having circumferential extending portions extending in the tire circumferential direction at positions differing in the tire width direction, a second lug groove being disposed between one center main groove of the two center main grooves and one shoulder main groove of the two shoulder main grooves which is adjacent to the one center main groove, the second lug groove extending in the tire width direction and having both ends opening to the one center main groove and the one shoulder main groove, a shoulder lug groove being disposed on the outer side in the tire width direction of the one shoulder main groove, the shoulder lug groove extending in the tire width direction across a ground contact edge and having one end opening to the one shoulder main groove, inclination directions of the second lug groove and the shoulder lug groove in the tire circumferential direction with respect to the tire width direction being mutually opposite directions, in a plurality of the second lug grooves disposed side by side in the tire circumferential direction, two types of the second lug grooves having different inclination angles with respect to the tire width direction being alternately disposed; the second lug groove opening to the one center main groove and the one shoulder main groove opening to the circumferential extending portions located close to the second lug groove among the circumferential extending portions comprised in respective of the one center main groove and the one shoulder main groove formed in the step shapes, the shoulder lug groove opening to the one shoulder main groove opening to the circumferential extending portion located close to the shoulder lug groove among the circumferential extending portions comprised in the one shoulder main groove formed in the step shape, the second lug groove comprising a first second lug groove and a second second lug groove having mutually different inclination angles with respect to the tire width direction and disposed alternately in the tire circumferential direction, the first second lug groove having a groove width that widens from an inner side in the tire width direction toward the outer side in the tire width direction, the second second lug groove having a groove width that widens from the outer side in the tire width direction toward the inner side in the tire width direction, and a relationship between a wavelength λ of the step shape of the main grooves and a pitch length P of the second lug groove being within a range $0.4 \leq (\lambda/P) \leq 0.6$, the pitch length P of the second lug groove being a distance in the tire circumferential direction between the first second lug grooves adjacent in the tire circumferential direction.

3. A tire, comprising: four main grooves extending in a tire circumferential direction; a plurality of land portions defined by the four main grooves; and the four main grooves comprising two center main grooves and two shoulder main grooves, the two center main grooves being disposed on both sides of a tire equatorial plane, the two shoulder main grooves being disposed on outer sides in a tire width direction of the two respective center main grooves, each of the four main grooves being formed in a step shape having circumferential extending portions extending in the tire circumferential direction at positions differing in the tire width direction, a second lug groove being disposed between one center main groove of the two center main grooves and one shoulder main groove of the two shoulder main grooves which is adjacent to the one center main groove, the second lug groove extending in the tire width direction and having both ends opening to the one center main groove and the one shoulder main groove, a shoulder lug groove being disposed on the outer side in the tire width direction of the one shoulder main groove, the shoulder lug groove extending in the tire width direction across a ground contact edge and having one end opening to the one shoulder main groove, inclination directions of the second lug groove and the shoulder lug groove in the tire circumferential direction with respect to the tire width direction being mutually opposite directions, in a plurality of the second lug grooves disposed side by side in the tire circumferential direction, two types of the second lug grooves having different inclination angles with respect to the tire width direction being alternately disposed; the second lug groove opening to the one center main groove and the one shoulder main groove opening to the circumferential extending portions located close to the second lug groove among the circumferential extending portions comprised in respective of the one center main groove and the one shoulder main groove

formed in the step shapes, the shoulder lug groove opening to the one shoulder main groove opening to the circumferential extending portion located close to the shoulder lug groove among the circumferential extending portions comprised in the one shoulder main groove formed in the step shape, no lug grooves extending in the tire width direction and having both ends opening to the center main groove and the shoulder main groove being disposed between circumferentially adjacent second lug grooves, the second lug groove comprising a first second lug groove and a second second lug groove having mutually different inclination angles with respect to the tire width direction and disposed alternately in the tire circumferential direction, and a relationship between a wavelength λ of the step shape of the main grooves and a pitch length P of the second lug groove being within a range $0.4 \leq (\lambda/P) \leq 0.6$, the pitch length P of the second lug groove being a distance in the tire circumferential direction between the first second lug grooves adjacent in the tire circumferential direction.

4. The tire according to claim 3, wherein the four main grooves have a width Ws in the tire width direction of a see-through portion as a portion through which it is seen within a range 1 mm<Ws≤4 mm in an extension direction view of the four main grooves.

5. The tire according to claim 3, wherein in the one center main groove and the one shoulder main groove, a relationship between a groove width Wce of the one center main groove and a groove width Wsh of the one shoulder main groove satisfies Wce≥Wsh.

6. The tire according to claim 3, wherein chamfered portions are formed at edges of the second lug groove and the shoulder lug groove, and the chamfered portion has a width We within a range 0.3 mm≤We≤0.8 mm and a depth De within a range 0.3 mm≤De≤0.8 mm.

7. The tire according to claim 3, wherein the second lug groove comprises a first second lug groove and a second second lug groove having mutually different inclination angles with respect to the tire width direction and disposed alternately in the tire circumferential direction, and the first second lug groove and the second second lug groove are inclined in the same direction with respect to the tire width direction.

8. The tire according to claim 3, wherein the second lug groove comprises a first second lug groove and a second second lug groove having mutually different inclination angles with respect to the tire width direction and disposed alternately in the tire circumferential direction, the first second lug groove has an angle $\theta\mathbf{1}$ with respect to the tire width direction within a range $15° < \theta\mathbf{1} \leq 35°$, the second lug groove has an angle $\theta\mathbf{2}$ with respect to the tire width direction within a range $25° \leq \theta\mathbf{2} \leq 55°$, and the angle $\theta\mathbf{1}$ of the first second lug groove and the angle $\theta\mathbf{2}$ of the second second lug groove with respect to the tire width direction satisfy $\theta\mathbf{1} < \theta\mathbf{2}$.

9. The tire according to claim 8, wherein the first second lug groove has a groove width that widens from an inner side in the tire width direction toward the outer side in the tire width direction, and the second second lug groove has a groove width that widens from the outer side in the tire width direction toward the inner side in the tire width direction.

10. The tire according to claim 8, wherein the first second lug groove has a groove depth that deepens from an inner side in the tire width direction toward the outer side in the tire width direction, and the second second lug groove has a groove depth that deepens from the outer side in the tire width direction toward the inner side in the tire width direction.

11. The tire according to claim 3, wherein a plurality of the land portions comprise a center land portion disposed between the two center main grooves, a second land portion disposed between the one center main groove and the one shoulder main groove adjacent in the tire width direction, and a shoulder land portion disposed on the outer side in the tire width direction of the one shoulder main groove, and in the center land portion, an open sipe having both ends opening to the two center main grooves is formed.

12. The tire according to claim 11, wherein semi-closed sipes are formed in the center land portion, the second land portion, and the shoulder land portion, and the semi-closed sipes each have one end opening to one of the four main grooves and another end terminating in one of the center land

portion, the second land portion, or the shoulder land portion.
13. The tire according to claim 12, wherein the semi-closed sipes open to the four main grooves with an inclination angle in the tire circumferential direction with respect to the tire width direction at an angle within a range of from 0° or more to 30° or less.
14. The tire according to claim 3, wherein the four main grooves have an offset amount Woff in the tire width direction between the circumferential extending portions having the different positions in the tire width direction within a range $0.3 \leq (Woff/Wgr) \leq 0.8$ with respect to a groove width Wgr of the four main grooves.
15. The tire according to claim 14, wherein the four main grooves have a width Ws in the tire width direction of a see-through portion as a portion through which it is seen within a range $1\ mm < Ws \leq 4\ mm$ in an extension direction view of the four main grooves.
16. The tire according to claim 15, wherein the second lug groove comprises a first second lug groove and a second second lug groove having mutually different inclination angles with respect to the tire width direction and disposed alternately in the tire circumferential direction, the first second lug groove has an angle $\theta\mathbf{1}$ with respect to the tire width direction within a range $15° < \theta\mathbf{1} \leq 35°$, the second second lug groove has an angle $\theta\mathbf{2}$ with respect to the tire width direction within a range $25° \leq \theta\mathbf{2} \leq 55°$, and the angle $\theta\mathbf{1}$ of the first second lug groove and the angle $\theta\mathbf{2}$ of the second second lug groove with respect to the tire width direction satisfy $\theta\mathbf{1} < \theta\mathbf{2}$.
17. The tire according to claim 16, wherein the first second lug groove has a groove width that widens from an inner side in the tire width direction toward the outer side in the tire width direction, and the second second lug groove has a groove width that widens from the outer side in the tire width direction toward the inner side in the tire width direction.
18. The tire according to claim 17, wherein the first second lug groove has a groove depth that deepens from an inner side in the tire width direction toward the outer side in the tire width direction, and the second second lug groove has a groove depth that deepens from the outer side in the tire width direction toward the inner side in the tire width direction.
19. The tire according to claim 18, wherein in the one center main groove and the one shoulder main groove, a relationship between a groove width Wce of the one center main groove and a groove width Wsh of the one shoulder main groove satisfies $Wce \geq Wsh$.
20. The tire according to claim 19, wherein chamfered portions are formed at edges of the second lug groove and the shoulder lug groove, and the chamfered portion has a width We within a range $0.3\ mm \leq We \leq 0.8\ mm$ and a depth De within a range $0.3\ mm \leq De \leq 0.8\ mm$.
21. The tire according to claim 20, wherein a plurality of the land portions comprise a center land portion disposed between the two center main grooves, a second land portion disposed between the one center main groove and the one shoulder main groove adjacent in the tire width direction, and a shoulder land portion disposed on the outer side in the tire width direction of the one shoulder main groove, and in the center land portion, an open sipe having both ends opening to the two center main grooves is formed.
22. The tire according to claim 21, wherein semi-closed sipes are formed in the center land portion, the second land portion, and the shoulder land portion, and the semi-closed sipes each have one end opening to one of the four main grooves and another end terminating in one of the center land portion, the second land portion, or the shoulder land portion.
23. The tire according to claim 22, wherein the semi-closed sipes open to the four main grooves with an inclination angle in the tire circumferential direction with respect to the tire width direction at an angle within a range of from 0° or more to 30° or less.